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Subject: Application for the Master Thesis "Development of a green reactor for wastewater treatment" (ID 52683) — Fraunhofer IGB, Stuttgart

Dear Ms. Munz,

I am a final-year Industrial Chemical Engineering student at INSAT (Université de Carthage, Tunisia), specializing in Process and Environment, and I am writing to apply for the master thesis "Development of a green reactor for wastewater treatment" at Fraunhofer IGB in Stuttgart. I am available from 1 February 2027, for 6 months (3 to 9 months possible), and I would carry out the work in English.

My training in water treatment and analytical chemistry rests on laboratory work. In an academic project on effluent treatment by coagulation-flocculation, I optimized the coagulation pH on a real effluent by testing 5 pH values and interpreted the result through aluminium speciation. In a project on the electrochemical treatment of an industrial effluent by electroflocculation, I removed 85% of Cd^{2+} and 57% of Zn^{2+} in 20 min and quantified the kinetics by pulse polarography. I also followed 5 anaerobic digestion configurations in ASBR digesters, reaching up to 0.50 L/d of biogas at 64.52% CH_4. Running experiments at laboratory scale and evaluating the data against the literature is exactly what your thesis requires.

Fraunhofer IGB's aim of demonstrating a new reactor at laboratory scale to set the target parameters of a pilot plant appeals to me, because it turns lab results directly into design decisions. Building a laboratory setup from scratch is new to me specifically, but my internships at ETAP, AGIL and Control Union, where I performed water, fuel and lubricant analyses under standardized procedures, show that I adapt quickly to a new laboratory and work independently.

I am drawn to this thesis because it sits precisely at the intersection of water treatment and process development.

I would welcome the opportunity to discuss my application further and am available for an interview at your convenience. Thank you for your consideration.

Sincerely,
Nermine Moumen